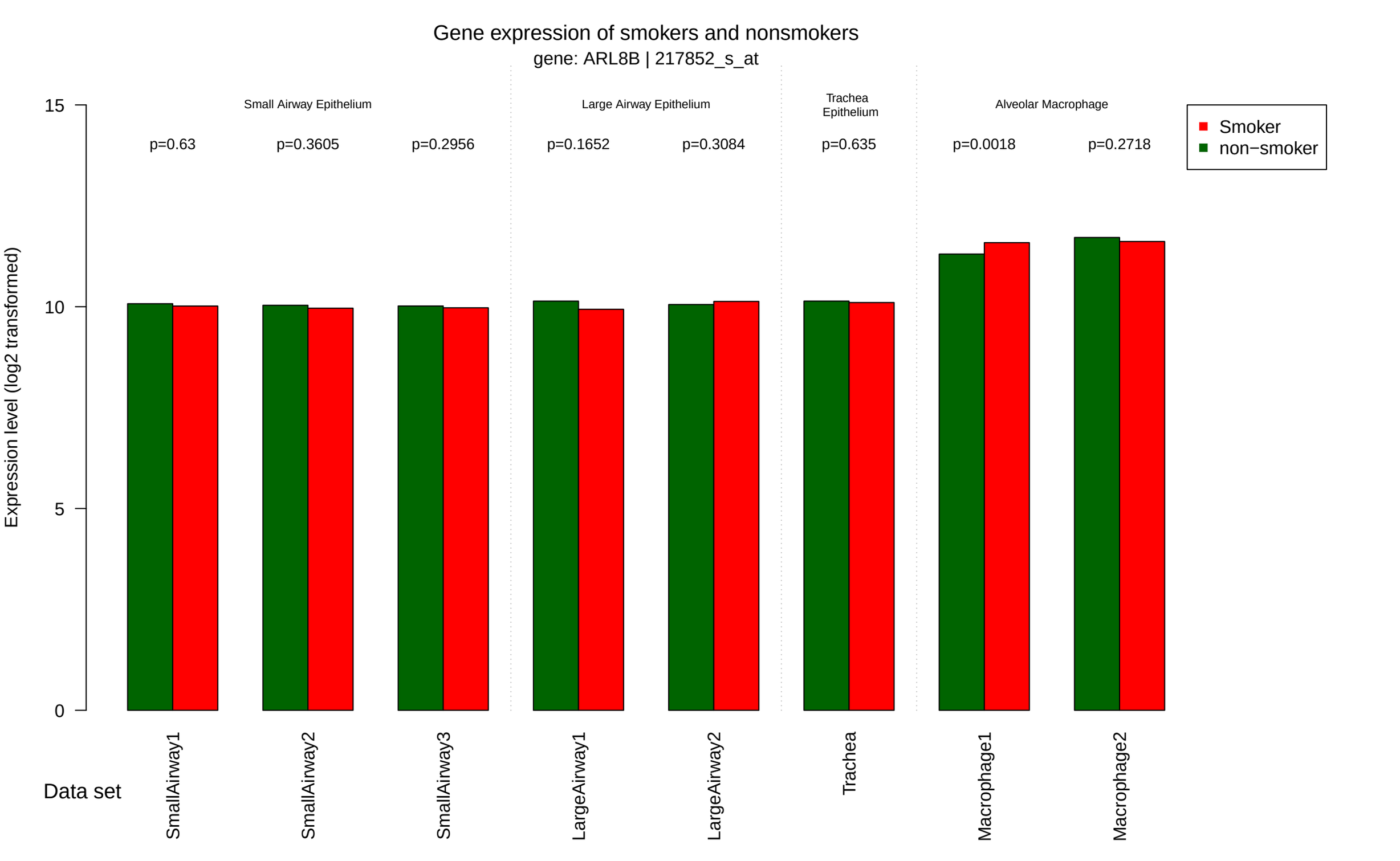
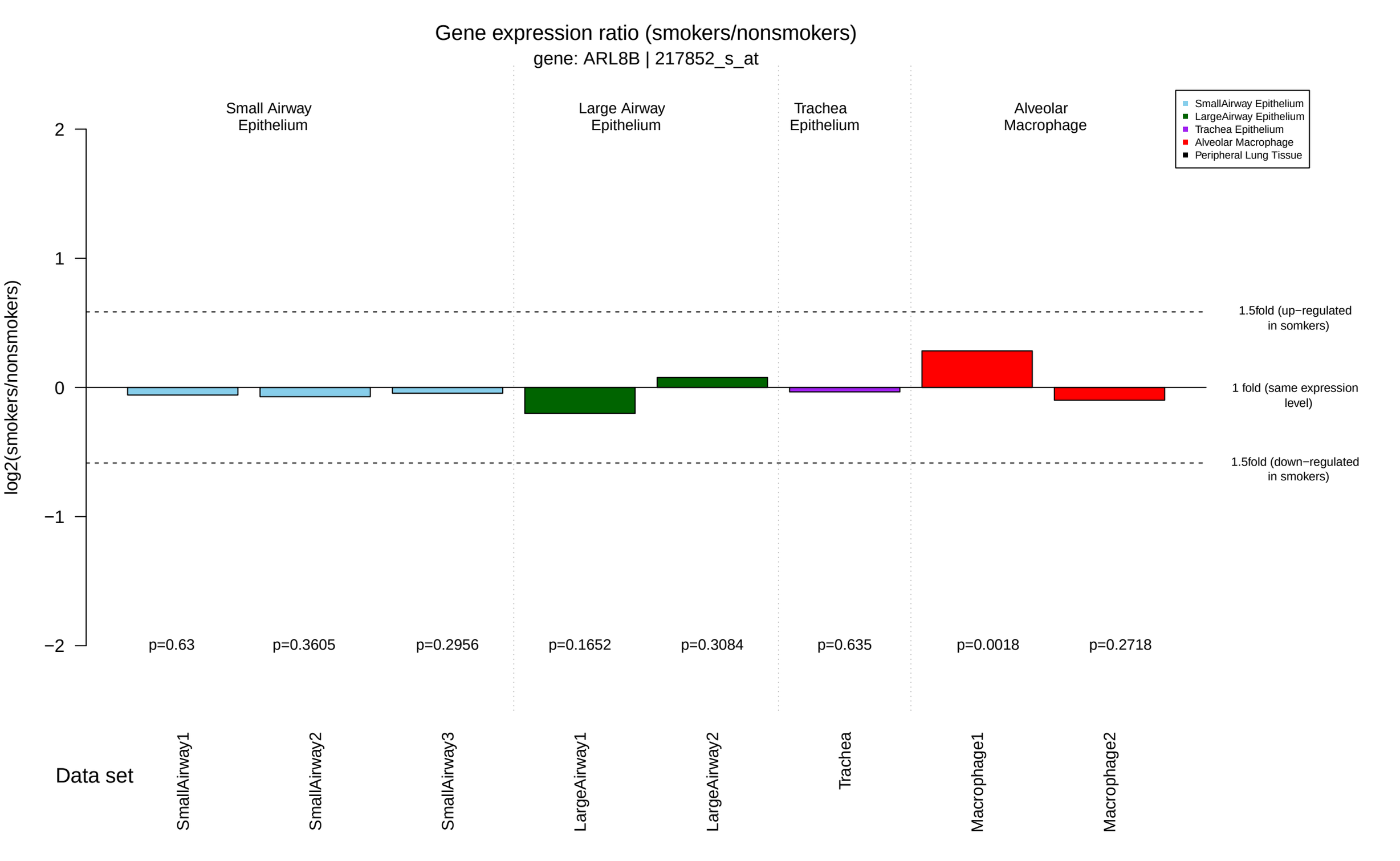
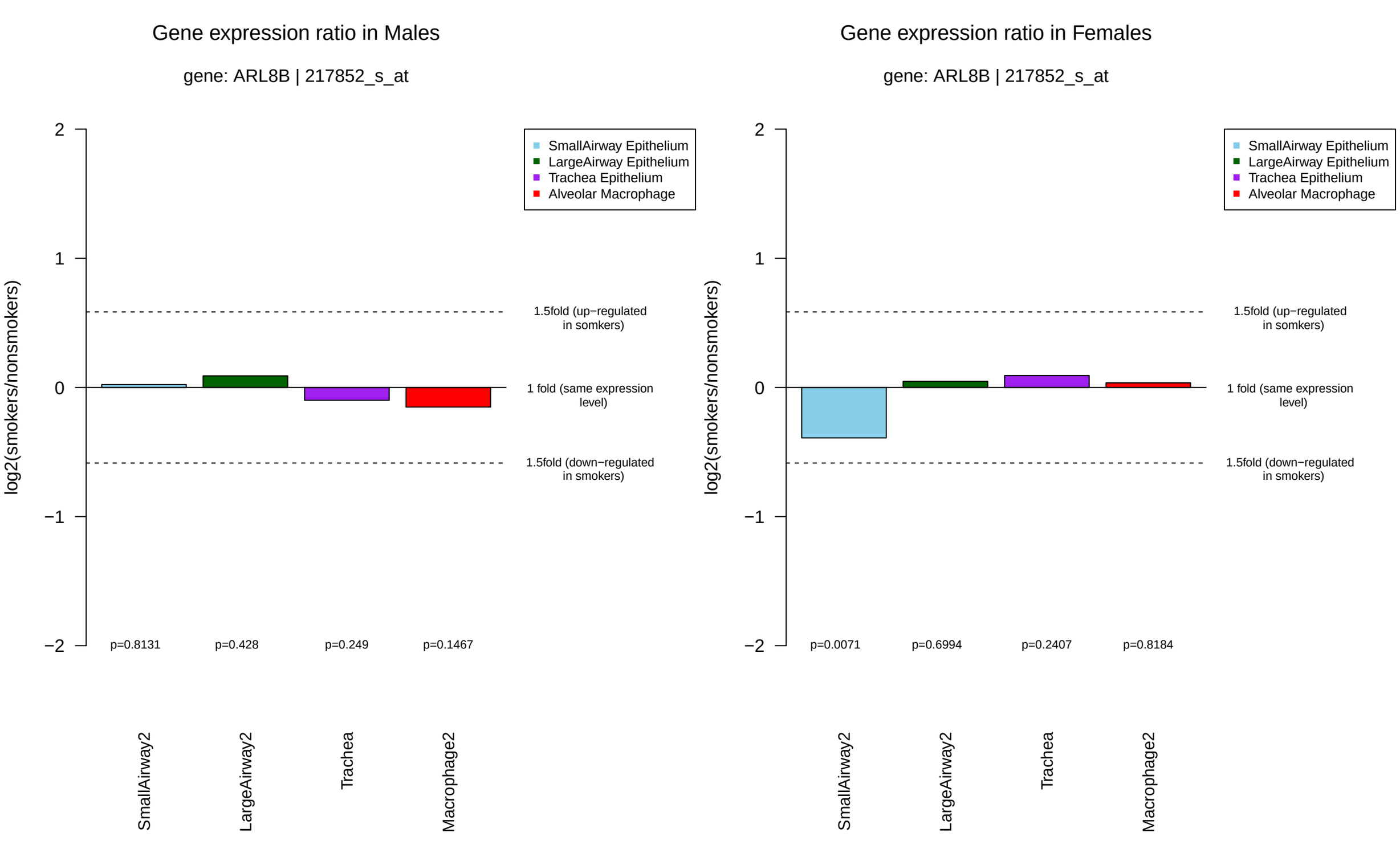
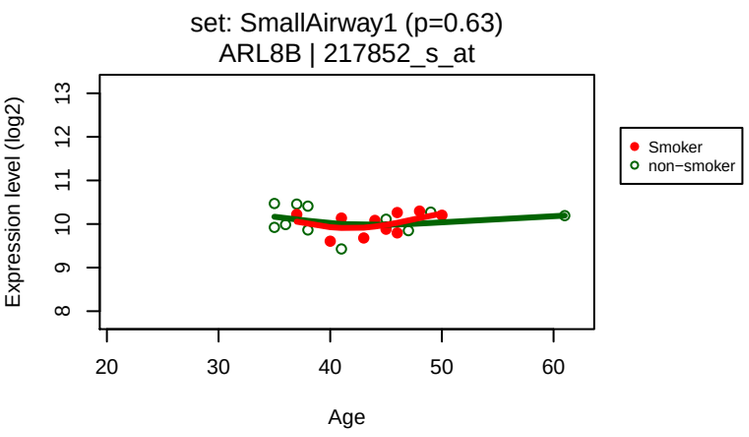
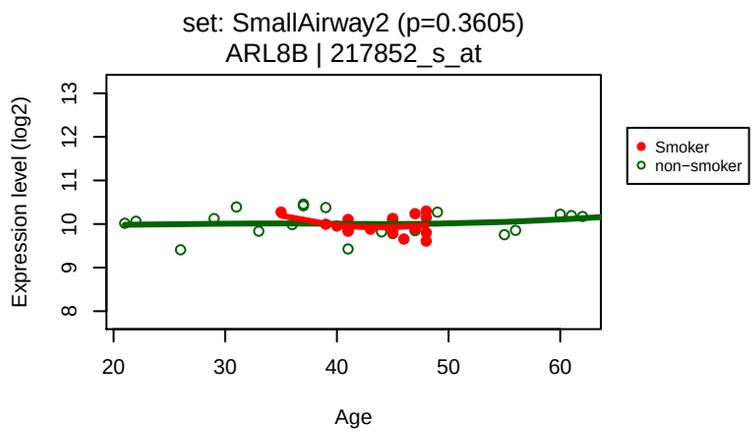
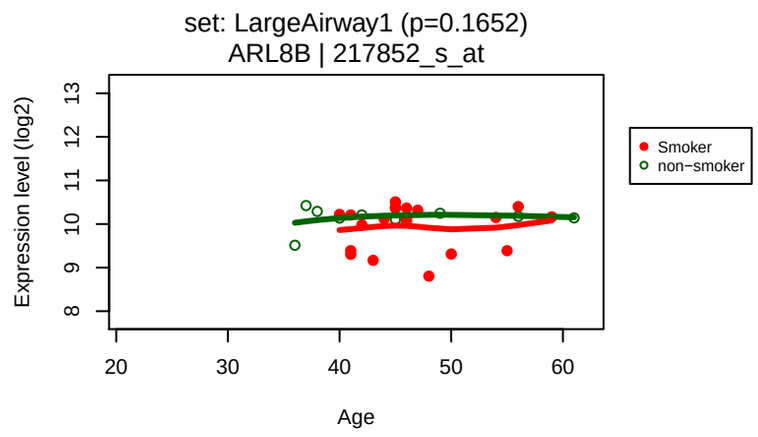

Gene expression of smokers and nonsmokers
gene: ARL8B | 217852_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.63
p=0.3605
p=0.2956
p=0.1652
p=0.3084
p=0.635
p=0.0018
p=0.2718
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

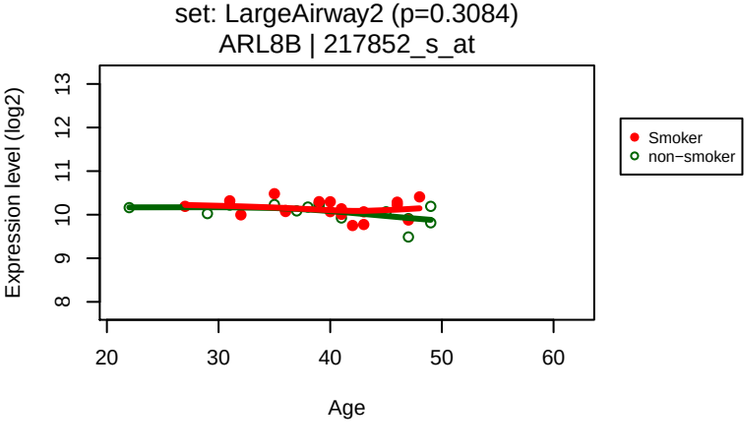

Gene expression ratio (smokers/nonsmokers)
gene: ARL8B | 217852_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.63
p=0.3605
p=0.2956
p=0.1652
p=0.3084
p=0.635
p=0.0018
p=0.2718
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

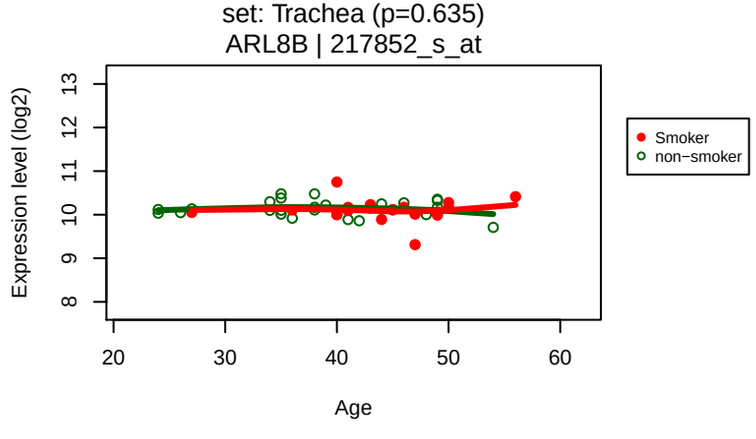

Gene expression ratio in Males
gene: ARL8B | 217852_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8131
p=0.428
p=0.249
p=0.1467
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL8B | 217852_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0071
p=0.6994
p=0.2407
p=0.8184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

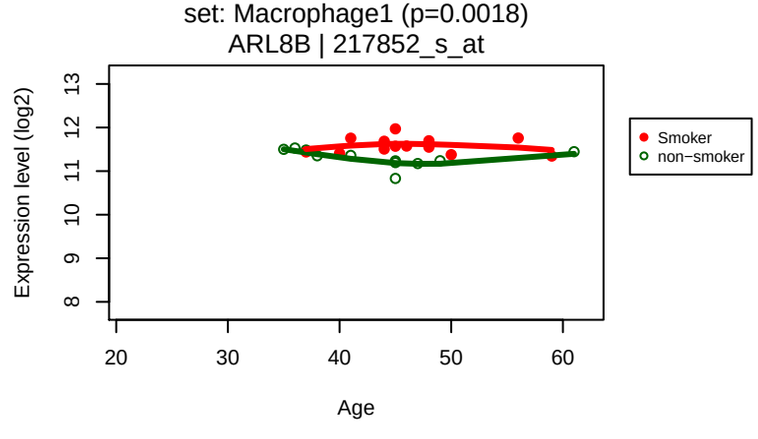

set: SmallAirway1 (p=0.63)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker

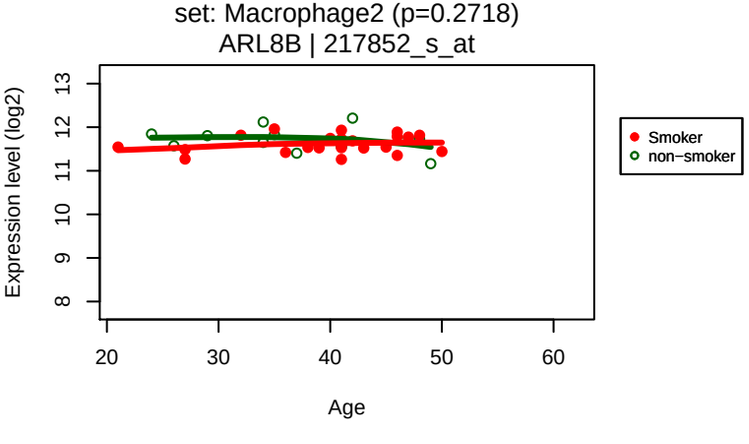

set: SmallAirway2 (p=0.3605)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker

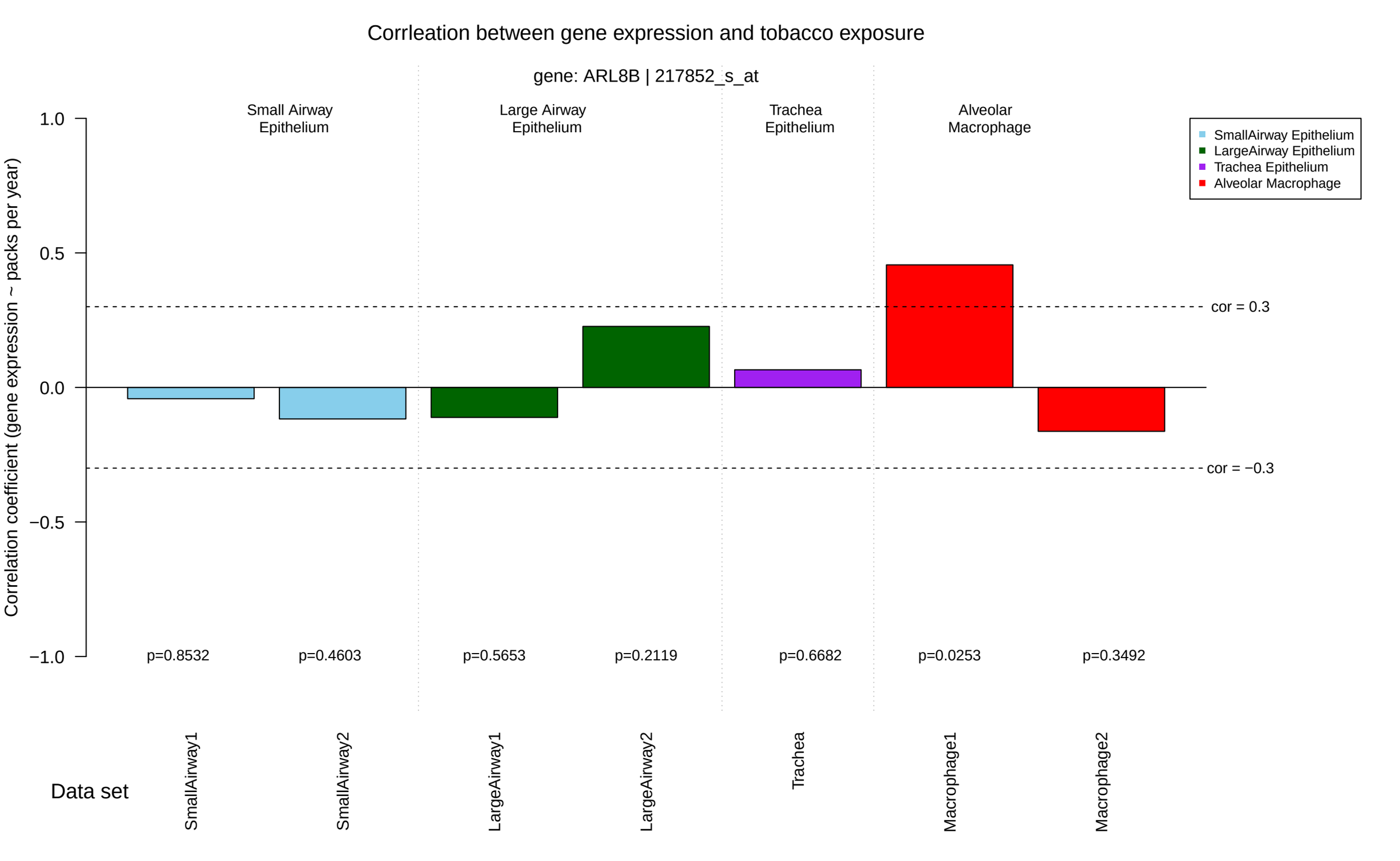

set: LargeAirway1 (p=0.1652)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3084)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.635)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0018)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2718)
ARL8B | 217852_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL8B | 217852_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8532
p=0.4603
p=0.5653
p=0.2119
p=0.6682
p=0.0253
p=0.3492
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2